

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 1.3: Quadratic Expressions and Equations — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.3: Quadratic Expressions and Equations
Driving Question	DRIVING QUESTION: "How can we use the mathematics of quadratic expressions and equations to predict, explain, and solve real problems — from the arc of a thrown ball on our school field to the dimensions of a farmer's shamba in Kericho?"

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Noticing the Curve — Introducing Quadratic Expressions Through the Shot Put Phenomenon	Learners observed a shot-put throw (live demonstration or video set on a school field) and plotted the ball's height against horizontal distance. They noted that the data points did not form a straight line — the curve rose, peaked, and fell symmetrically. When second differences of the height values were computed from a data table, they were constant (non-zero), signalling a quadratic rather than a linear relationship. Teacher should display both a linear and a quadratic data set side-by-side so learners can contrast first-difference (linear) vs. second-difference (quadratic) patterns before naming the rule.	Learners learned that a quadratic expression contains an x^2 term as its highest power, and that this structure is necessary to model any relationship whose second differences are constant. They were introduced to the general form $ax^2 + bx + c$ ($a \neq 0$) and identified a , b , and c in expressions drawn from the shot-put context (e.g., $h = -x^2 + 6x + 1$). They also learned that setting $a = 0$ collapses the expression to a linear one, reinforcing why the x^2 term is the defining feature. Pedagogical note: Emphasise that 'quadratic' comes from the Latin quadratus (square) — the x^2 term literally represents a square area, linking naturally to the shamba context introduced in Lesson 2.	Learners explained that the curved arc of the shot put cannot be described by a straight-line (linear) expression because the rate of change is itself changing. The constant second difference is the mathematical fingerprint of a quadratic relationship. By identifying a , b , and c in $h = -x^2 + 6x + 1$, learners connected the negative leading coefficient ($a < 0$) to the fact that the parabola opens downward — matching the physical reality that the ball eventually falls. Teacher should cold-call at least three learners to state: 'The expression is quadratic because... and the parabola opens [upward/downward] because...' before moving on.
Lesson 2 Forming Quadratic Expressions: From the Shot Put Arc to the Shamba	Learners observed a diagram of a rectangular shamba in Kericho whose length is $(x + 4)$ metres and width is x metres. They computed numerical areas for several values of x (e.g., $x = 2, 3, 5, 10$) and recorded results in a table. They then plotted Area vs. x and noted the same upward-curving shape seen in the shot-put	Learners learned that multiplying two linear expressions — $(x)(x + 4)$ — produces $x^2 + 4x$, a quadratic expression, because the $x \times x$ product generates the x^2 term. They practised forming quadratic expressions from other rectangular contexts (e.g., a maize plot of width x and length $2x + 3$ giving $2x^2 + 3x$) and identified that the	Learners explained that a quadratic expression arises naturally whenever area is computed as the product of two variable lengths — the same algebraic structure underlies both the shamba problem and the shot-put arc. They articulated that $x^2 + 4x$ represents the total area of the shamba and that each term has a geometric meaning: x^2

	arc. Second differences of the area values were again constant, confirming the quadratic nature. Teacher should circulate during table-completion and ask: 'What do you notice about how quickly the area grows as x increases?' — allow 30 seconds wait time before taking responses.	coefficient of x^2 equals the product of the leading coefficients of the two linear factors. They also revisited second differences as a verification tool: if second differences of the area sequence are constant, the expression is quadratic. Pedagogical note: Have learners write the multiplication as a two-row table (grid method preview) so the connection to Lesson 6 (FOIL/box method) is foreshadowed.	is the square portion and $4x$ is the rectangular strip. Teacher should ask: 'If the farmer in Kericho wants to enclose exactly 60 m^2 with these dimensions, what equation must she solve?' — this question deliberately plants the seed for equation-solving in Lessons 7–8 without requiring an answer yet. Record learner responses on the Driving Question Board.
Lesson 3 Factorising Quadratic Expressions — Breaking the Curve's Code	Learners observed a worked example on the board: the expansion $(x + 3)(x + 2) = x^2 + 5x + 6$ was displayed, then the arrow was reversed — given $x^2 + 5x + 6$, can we recover the two brackets? Learners attempted to find two integers whose product is 6 and sum is 5, using a factor-pair table. They also observed the ac method applied to $2x^2 + 7x + 3$ ($a \neq 1$ case): multiply $a \times c = 6$, find pairs summing to 7 (1 and 6), split the middle term, then group. Teacher should display both methods simultaneously and ask learners to vote (hands or slates) on which method they find more transparent before proceeding.	Learners learned that factorising a quadratic expression reverses the expansion process. For the $a = 1$ case ($x^2 + bx + c$), they find two integers p and q such that $p \times q = c$ and $p + q = b$, then write $(x + p)(x + q)$. For the $a \neq 1$ case, they apply the ac method: compute ac , find a factor pair summing to b , split the middle term, and factor by grouping. They learned that not all quadratics factorise over the integers (e.g., $x^2 + x + 1$) and that checking by re-expanding is essential. Pedagogical note: Insist learners verify every factorisation by expanding back — frame this as 'the reverse journey back to the shamba area' to maintain contextual coherence.	Learners explained that the two factors of a quadratic expression represent the two linear dimensions whose product gives the area (or any quadratic quantity in context). For example, $x^2 + 5x + 6 = (x + 3)(x + 2)$ means a shamba of area $(x^2 + 5x + 6) \text{ m}^2$ has dimensions $(x + 3) \text{ m}$ and $(x + 2) \text{ m}$. They connected factorisation to the driving question by noting that finding the factors is the first step toward finding the value of x that makes the area equal a specific target — setting up the concept of a quadratic equation. Teacher should cold-call three learners: 'State the factor pair and explain why it works using both the product AND the sum condition.'
Lesson 4 Completing the Square and the Difference of Two Squares	Learners observed two special quadratic structures side-by-side on the board: (1) $x^2 - 9$, presented as the difference of two perfect squares, and (2) $x^2 + 6x + \underline{\hspace{1cm}}$, presented as an incomplete perfect square. For (1), they were shown a geometric square of side x with a smaller square of side 3 removed from its corner, illustrating $(x - 3)(x + 3)$ visually. For (2), they completed a square diagram by adding a $(b/2)^2$ tile. Teacher should ask learners to predict — before any instruction — what integer fills the blank in $x^2 + 6x + \underline{\hspace{1cm}}$ to make a perfect square, and collect predictions on the board before revealing the answer.	Learners learned two special factorisation patterns: the Difference of Two Squares, $a^2 - b^2 = (a - b)(a + b)$, and the Perfect Square Trinomial, $a^2 + 2ab + b^2 = (a + b)^2$. They practised recognising these patterns in expressions such as $4x^2 - 25$ and $x^2 + 10x + 25$. They also learned the completing-the-square technique: rewriting $x^2 + bx + c$ as $(x + b/2)^2 - (b/2)^2 + c$, which expresses the quadratic in vertex form and reveals the minimum or maximum value. Pedagogical note: Stress that completing the square is not merely a trick — it is the algebraic equivalent of the geometric square-completion diagram, and it will later be used to derive the quadratic formula.	Learners explained how recognising the Difference of Two Squares pattern dramatically shortens factorisation: instead of searching for a factor pair, they simply identify a and b from $a^2 - b^2$ and write the two conjugate factors immediately. They also explained that completing the square on the shot-put height expression $h = -x^2 + 6x + 1$ (rewritten as $-(x - 3)^2 + 10$) reveals the maximum height of 10 metres at $x = 3$ metres — a real, physically meaningful result. Teacher should connect this back to the Driving Question Board: 'Which of our earlier questions does knowing the maximum height answer?' Allow learners to update their DQB entries.
Lesson 5 Factorising Quadratic Expressions —	Learners observed a set of six quadratic expressions derived from earlier shamba and shot-put contexts and sorted them into	Learners learned that every quadratic expression of the form $x^2 + bx + c$ can be written as a product of two brackets $(x + p)$	Learners explained how the structure of the factors predicts the x -intercepts of the parabola: the values of x that make each

Unlocking the Hidden Structure of the Curve	three categories: (a) factorisable over the integers, (b) perfect square trinomials, and (c) difference of two squares. The sorting activity was done in pairs using printed cards. Learners then examined the graphical behaviour of each type — where the parabola crosses the x-axis (two points, one point tangent, or no real crossings) — using a pre-drawn set of parabola sketches. Teacher should allow 2 minutes of silent individual sorting before pair discussion, then cold-call three pairs to justify their categorisation.	$(x + q)$ where $p \times q = c$ and $p + q = b$, consolidating the product-sum rule. They deepened understanding by connecting the signs of p and q to the signs of b and c: if $c > 0$ and $b > 0$, both factors are positive; if $c > 0$ and $b < 0$, both are negative; if $c < 0$, one factor is positive and one negative. Learners also revisited the $a \neq 1$ case using the ac method and identified perfect-square and difference-of-squares expressions as efficient special cases. Pedagogical note: Use a colour-coding system (e.g., green for positive, red for negative) on the board to help learners track sign rules without memorising them as isolated facts.	bracket zero are the points where the curve crosses the x-axis (the roots). For example, $(x + 3)(x - 2) = 0$ gives roots $x = -3$ and $x = 2$, meaning the shot put (if modelled by this expression) lands at $x = 2$ metres and was launched from $x = -3$ (a negative value flagged as contextually impossible, prompting discussion about domain). Teacher should ask: 'If a shamba's area is modelled by $x^2 + x - 6$, what are the possible widths of the shamba, and which values of x make physical sense?' — use this to preview equation-solving in Lesson 7.
Lesson 6 Expanding Products of Two Linear Brackets: Grid/Box Method and FOIL in Kenyan Contexts	Learners observed the teacher expanding $(x + 5)(x + 3)$ using three representations simultaneously: (1) the FOIL mnemonic labelled on the brackets, (2) a 2×2 grid/box diagram with each term in its own cell, and (3) a geometric rectangle partitioned into four sub-areas on squared paper. All three yielded $x^2 + 8x + 15$. Learners then expanded $(2x + 3)(x - 4)$ using the grid method independently and compared answers with a partner. Teacher should circulate and specifically check that learners correctly handle the negative sign in $(x - 4)$: a common error is writing $+3 \times (-4) = +12$ — address this publicly if observed.	Learners learned that the product of two linear expressions $(ax + b)(cx + d)$ always produces a quadratic expression with three terms: $acx^2 + (ad + bc)x + bd$. The grid/box method organises this multiplication so each of the four partial products occupies its own cell, reducing sign errors. FOIL (First, Outer, Inner, Last) provides the same four products in a sequential order. Learners practised both methods using Kenyan contexts: the cost of fencing a shamba of length $(2x + 1)$ m at a cost of $(x + 3)$ Ksh per metre, and the total mass of ugali portions of size $(x + 2)(x - 1)$ grams. Pedagogical note: Require learners to always write all four partial products before collecting like terms — do not allow 'shortcutting' until the process is automatic.	Learners explained that expansion and factorisation are inverse operations: expanding $(x + 5)(x + 3)$ gives $x^2 + 8x + 15$, and factorising $x^2 + 8x + 15$ recovers $(x + 5)(x + 3)$. This reversibility is the key insight underpinning the entire sub-strand. They connected the grid method back to the shamba area diagram: each cell of the grid corresponds to a rectangular sub-plot of the shamba, so the algebraic and geometric representations are identical. Teacher should ask learners to write one sentence on their exercise books: 'Expanding and factorising are related because...' and share with a neighbour before two learners read aloud — reinforcing metacognitive articulation of the inverse relationship.
Lesson 7 Forming and Solving Quadratic Equations in Real-Life Situations	Learners observed three word problems projected on the board, all set in Kenyan contexts: (1) A rectangular shamba in Kericho has length 5 m more than its width x; its area is 84 m^2 — find x. (2) A ball is thrown upward from a height of 1 m on the school field; its height after t seconds is $h = -t^2 + 6t + 1$ — when does it hit the ground? (3) A sukuma wiki bed has perimeter 26 m and area 40 m^2 — find its dimensions. Learners first wrote the quadratic equation in standard form for each problem before attempting any solution. Teacher should	Learners learned to form quadratic equations in standard form $(ax^2 + bx + c = 0)$ from real-life word problems involving area, projectile motion, and perimeter-area pairs. They solved these equations by: (a) extracting a common factor where applicable (e.g., $2x^2 + 6x = 0 \rightarrow 2x(x + 3) = 0$), and (b) trinomial factorisation (e.g., $x^2 + 5x - 84 = 0 \rightarrow (x + 12)(x - 7) = 0$). They learned to apply the Zero Product Property — if $AB = 0$, then $A = 0$ or $B = 0$ — and to reject solutions that are contextually invalid (e.g., negative lengths). Pedagogical note:	Learners explained that solving a quadratic equation by factorisation means finding the values of the unknown that make the expression equal to zero — i.e., the roots of the equation. In the shamba context, $x = 7$ (rejecting $x = -12$) gives a width of 7 m and a length of 12 m, which multiplies to 84 m^2 — confirming the solution. In the projectile context, $-t^2 + 6t + 1 = 0$ identifies the landing time, answering the driving question about the ball's path. Teacher should return to the Driving Question Board and ask: 'Which questions on our DQB can we now

	give 3 minutes of silent individual work on equation-formation only, then cold-call three learners to write their equations on the board for class verification.	Explicitly model the three-step solution frame: FORM the equation → FACTORISE → SOLVE and CHECK. Require learners to write all three steps for every problem.	fully answer?' Facilitate a class tick-off of resolved questions and note which questions remain open for Lesson 8.
Lesson 8 Final Explanation: Reversing the Expansion — Factorisation, Difference of Two Squares, and the Full Picture of the Parabolic Path	Learners observed a comprehensive summary display showing the full journey of the sub-strand: the shot-put arc photograph → the data table with constant second differences → the quadratic expression → expansion by FOIL/grid → factorisation (trinomial, difference of squares, perfect square) → equation in standard form → solution by factorisation → contextual interpretation. Each stage was mapped onto the driving question. Learners also observed a final 'gallery walk' of six posters (one per group) showing worked solutions to problems from Lessons 1–7, annotated with the mathematical technique used. Teacher should allocate 8 minutes for silent gallery viewing before whole-class debrief.	Learners learned that factorisation is the reverse of expansion: every quadratic expression arising from an area context was originally a product of two linear factors (the dimensions), and factorisation recovers those dimensions. They consolidated all factorisation techniques: (a) common factor, (b) difference of two squares $a^2 - b^2 = (a - b)(a + b)$, (c) perfect square trinomial, and (d) general trinomial factorisation ($a = 1$ and $a \neq 1$ cases). They also revisited completing the square as an alternative route to solving equations and identifying the vertex. The full conceptual map — expression → equation → solution → interpretation — was explicitly articulated and recorded. Pedagogical note: Use the final lesson to address any persistent misconceptions identified in earlier lessons (e.g., sign errors in ac method, forgetting to reject negative roots in context) by displaying anonymised student errors from prior lessons and asking the class to diagnose and correct them.	Learners explained the complete mathematical narrative: the parabolic arc of a shot put and the area of a Kericho shamba are both modelled by quadratic expressions because each arises from a product of two variable quantities. Expanding reveals the trinomial form; factorising reverses it; setting the expression equal to zero and solving identifies the critical real-world values (landing point, required dimension). The Difference of Two Squares pattern (e.g., $x^2 - 16 = (x - 4)(x + 4)$) shows that some shamba configurations have symmetric solutions. Learners presented their final cumulative models — updated diagrams linking every lesson's key idea — to the class, demonstrating mastery of the full sub-strand and providing a complete answer to the driving question. Teacher should close by asking three learners to each complete the sentence: 'We can use quadratic expressions and equations to predict and solve real problems because...'